

Practical: AI Ethics and Governance

From the from the previous lesson we had learned about what is AI ethics and Governance. We had introduce some concept of Explainer AI methods and tool. Let look at some of these methods and tools.

Section 1: Visual Inspection – Image Classification

<http://www.heatmapping.org/>

Start the link in the browser, then select the following in RED

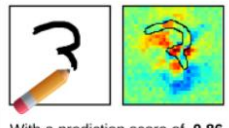
Not secure | heatmapping.org

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aspects, i.e., how to best include interpretation methods into the standard usage of ML; and 4) demonstrate successful usage of XAI in a representative selection of application scenarios. Finally, we discuss challenges and possible future directions of this exciting foundational field of ML.

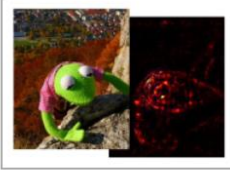
Interactive LRP Demos

Draw a handwritten digit and see the heatmap being formed in real-time. Create your own heatmap for natural images or text. These demos are based on the Layer-wise Relevance Propagation (LRP) technique by Bach et al. (2015).



With a prediction score of **0.86** the digit was classified as **3**

MNIST: A simple LRP demo based on a neural network that predicts handwritten digits and was trained using the MNIST data set.



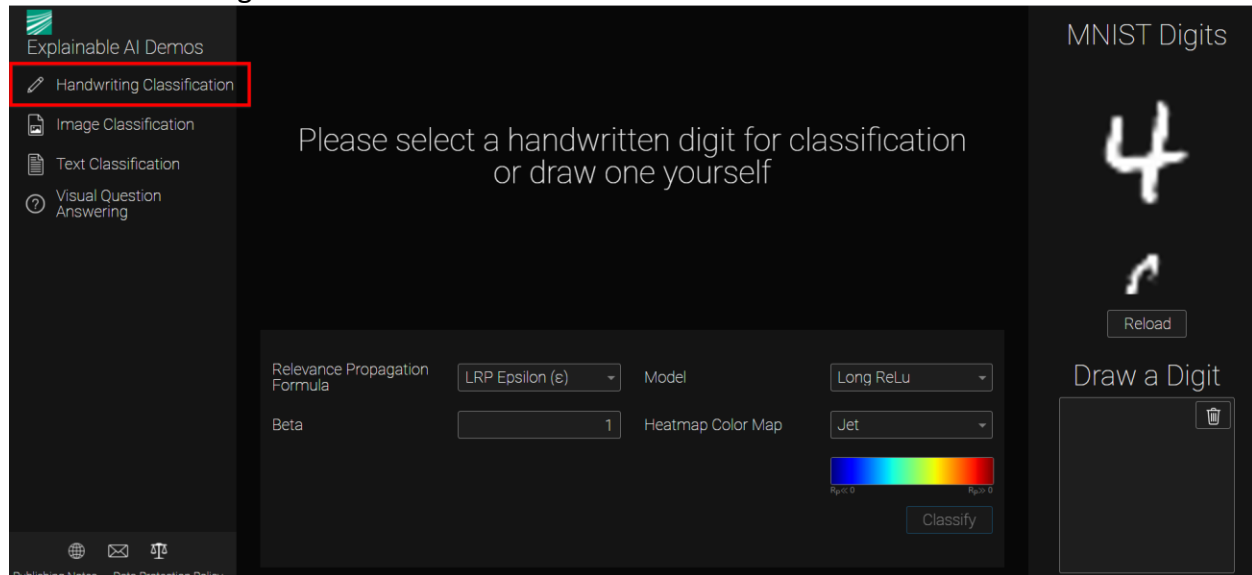
Caffe: A more complex LRP demo based on a neural network implemented using Caffe. The neural network predicts the contents of the picture.

Highest Prediction Scores		Most Relevant Words	
Class	Prediction Score	Word	Relevance Value
comp.graphics	7.8762	file	0.5385
comp.os.ms-windows.misc	5.8869	programs	0.4320
comp.windows.x	4.6542	pics	0.7503
		qqa	1.3609

> I've got this **file** with a .dl extension.
> what is this? Are there any 40-200 or 20-2 programs
> what will run this file? Thanks.

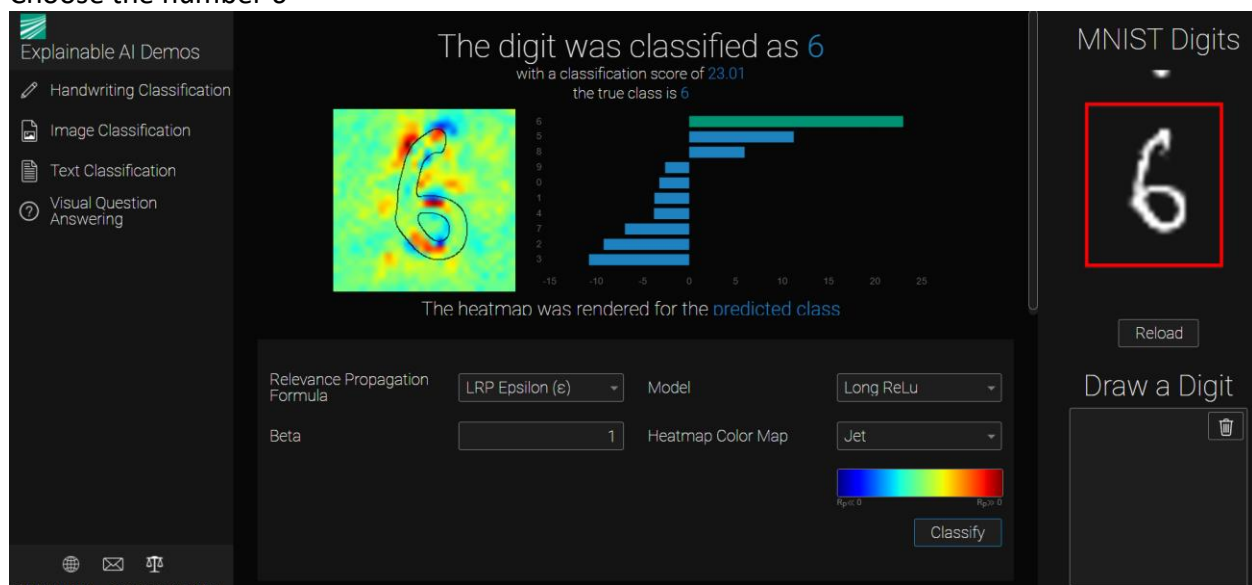
Text: A LRP demo that explains classification on natural language. The neural network predicts the type of document.

Select Handwriting Classification.



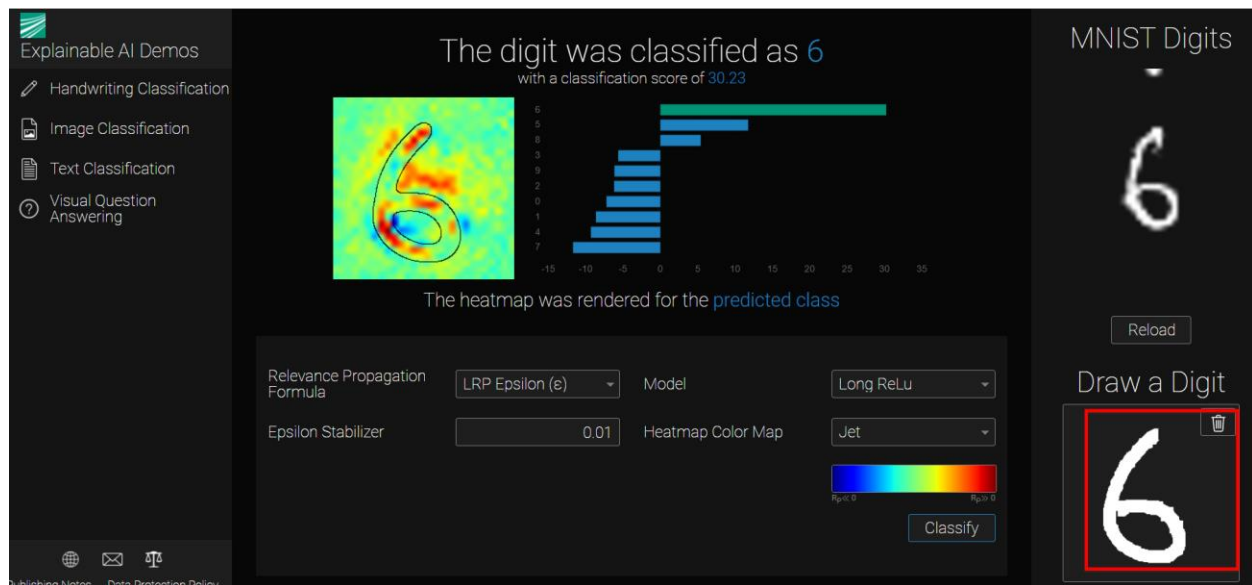
Exercise 1

Choose the number 6

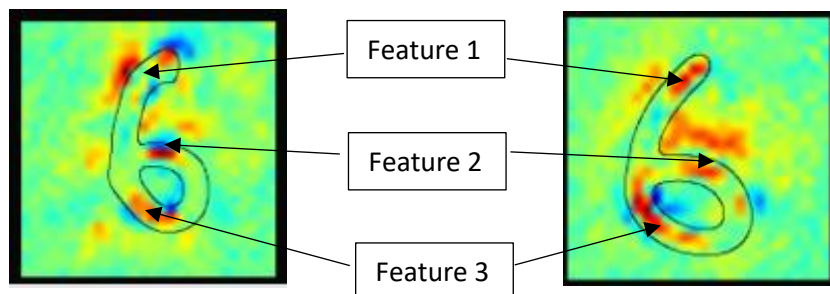


Observe the heatmap of the number 6. See the the red color location in the heatmap.

Then draw your number 6 at the box Draw a Digit. See the red color location in the heatmap.



Compare the 2 heatmap for the 2 different handwriting of 6.



Both of the handwriting 6 are different images but the trained model seem using the similar features to detect the image as number 6.

Through the heatmap showing the activation points where the trained model used to predict the number 6.

In this visual display it allowed us to explain how the Deep learning model trained to predict the different handwriting image.

Try to repeat the above for different handwriting number. Identify the area of the heatmap where it used to predict the handwriting number in the image.

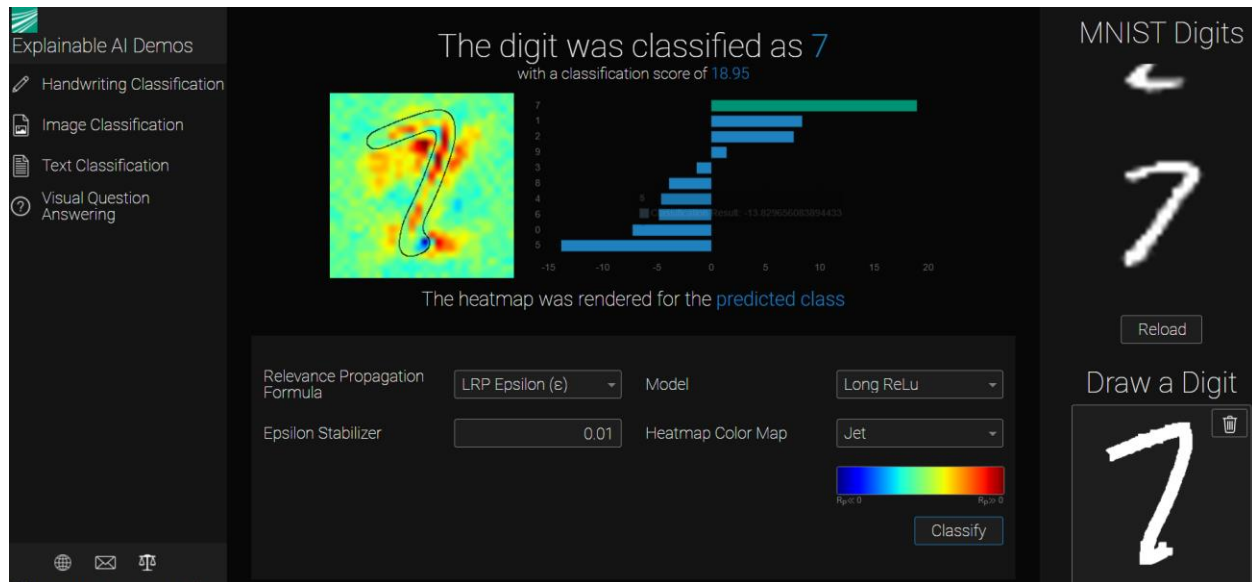
Exercise 2

Now try to compare the heatmap for the handwriting 2 and 7.



Notice both the headmap at the red region are quite similar except for the number 2. At the end there is horizontal extension to form a 2.

Try to wrote you own 2 with a shorter horizontal extension and see the heatmap and predicted result.

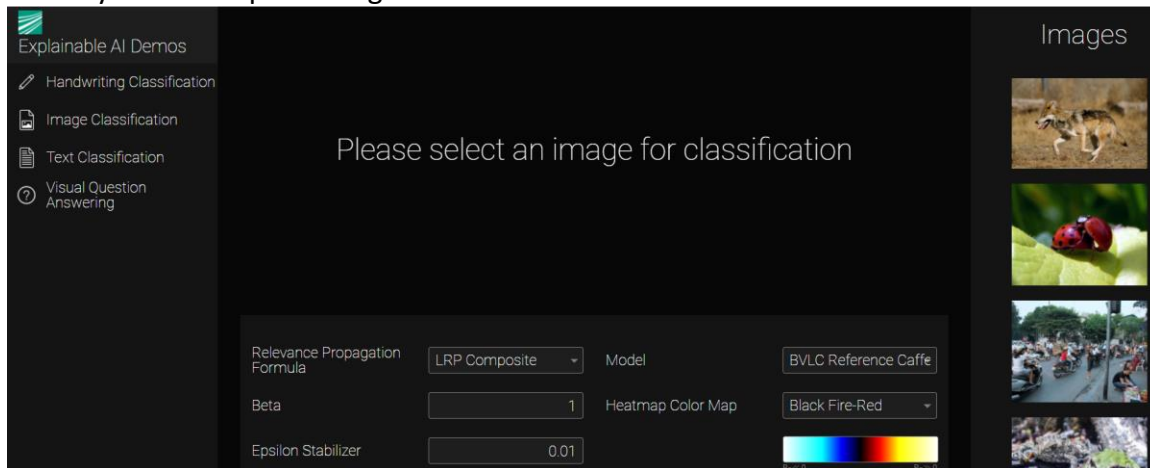


You notice the predicted result is 7 instead. How to you explain why there is an different result from the expected handwriting number 2?

We had observed most of the region that used to identify 2 and 7 are quite similar. But the important region to distinct 2 amd 7 is the horizontal line the end. In this particular handwriting number 2, the horizontal line at the end is too short to be identify are the 2, so the model is predict it to the closest number 7 based on the headmap region.

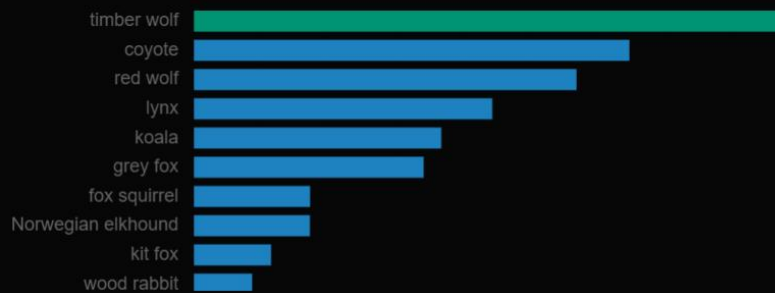
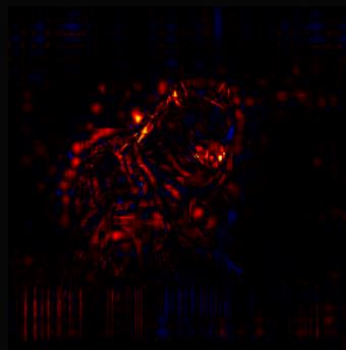
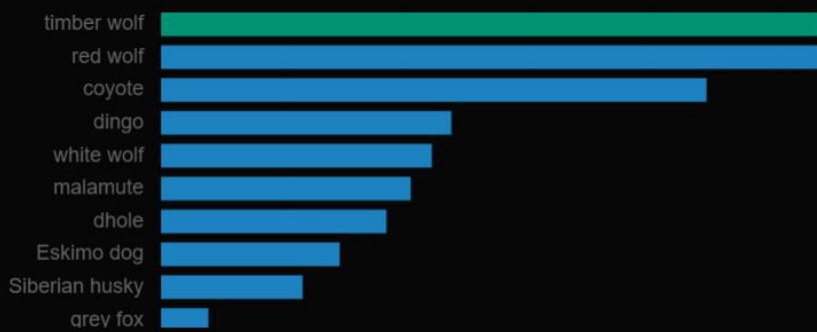
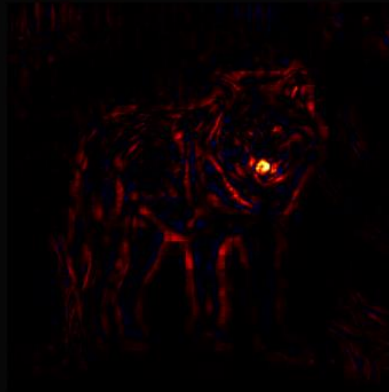
Exercise 3

Now try more complex image classification.

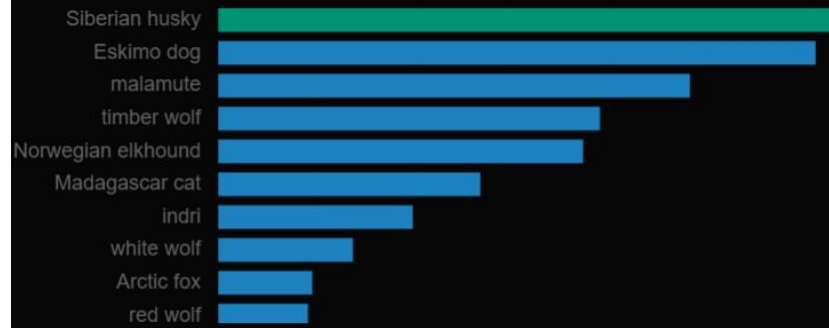
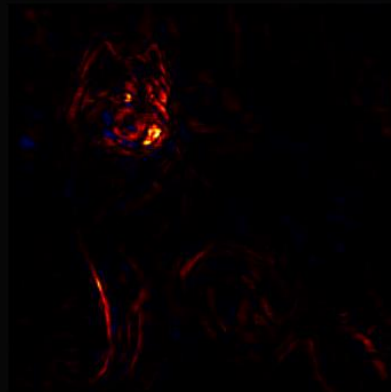


Test image with the dog1.jpg, dog2.jpg, wolf1.jpg and wolf2.jpg.

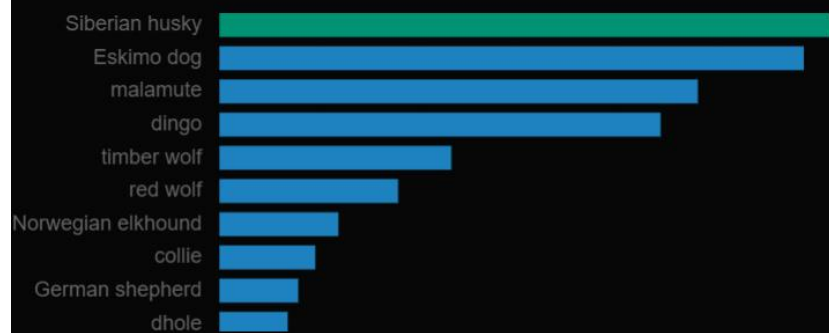
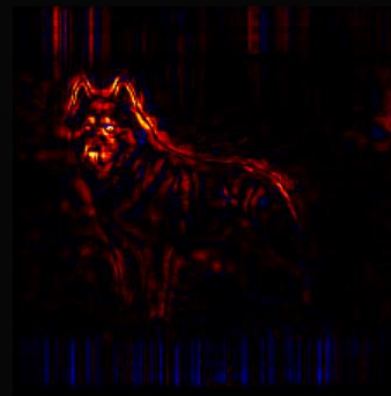
with a classification score of 34.80



with a classification score of 17.30



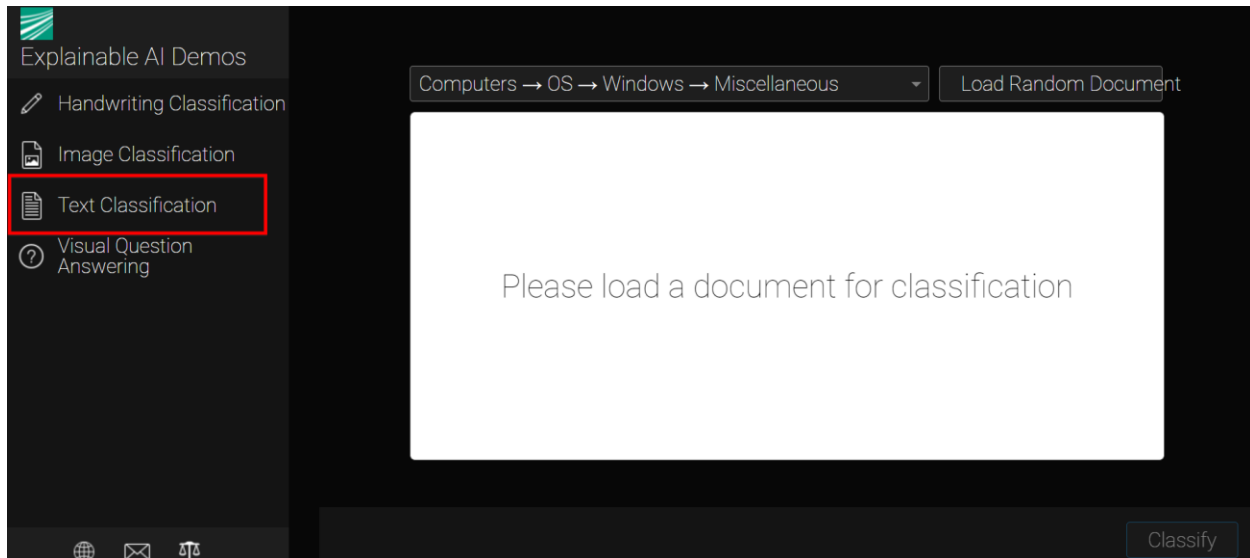
with a classification score of 17.16



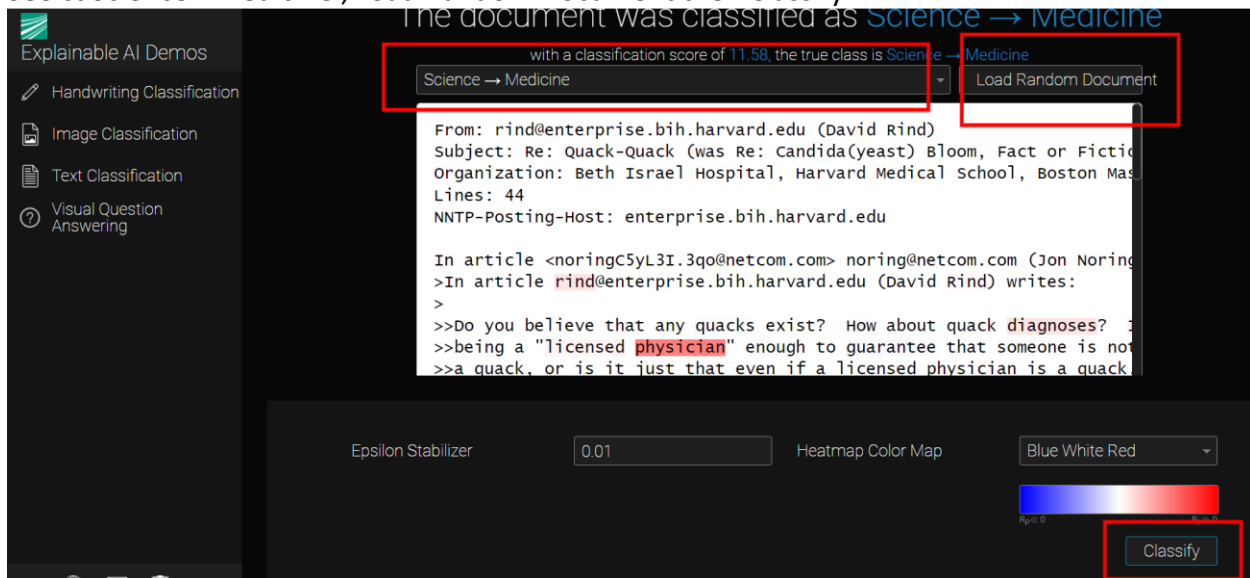
Can you explain how the model classify the 2 different type of animals?

Section 2: Visual Inspection – Text Classification

Select the Text Classification



Select Science->Medicine , Load Random Document then Classify



See the highlighted word in Red to check it help to identify the topic for Science->Medicine?

Can you identify the importance words used to explain why it should be a Medicine document?

Science → Medicine
Load Random Document

NNTP-Posting-Host: enterprise.bih.harvard.edu

In article <noringC5yl3I.3qo@netcom.com> noring@netcom.com (Jon Noring) writes:
>In article rind@enterprise.bih.harvard.edu (David Rind) writes:
>
>>Do you believe that any quacks exist? How about quack diagnoses? Is
>>being a "licensed physician" enough to guarantee that someone is not
>>a quack, or is it just that even if a licensed physician is a quack,
>>other people shouldn't say so?

>I would say there are also significant numbers of unscrupulous doctors (of
>the squeaky-clean, traditional crew-cut, talk to the AMA before starting
>any treatment, kind)

Umm, weren't you the one objecting to someone who is a "licensed
physician" being called a quack? Or is it just that being a licensed
physician is a good defense against charges of quackery when the
physician agrees with your system of beliefs?

>Lately I've seen the word "quack" bandied about recklessly.

Actually, I almost never use the term quack. When I discuss
"systemic yeast syndrome", however, I always point out that
mainstream medicine views this as a quack diagnosis (and I agree
with that characterization).

>Let me put it another way to make my point clear: "quack" is a nebulous word
>lacking in any precision.

Science → Medicine
Load Random Document

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"systemic yeast syndrome", however, I always point out that
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>lacking in any precision.

Really? I bet virtually everyone reading these posts understands what
Steve Dyer, Gordon Banks, and I am implying when we have talked about
systemic yeast syndrome as a quack diagnosis. Would you prefer the
word "charlatan"? (I don't happen to think that all quacks are
charlatans since I suspect that some believe in the "diseases" they
are diagnosing.)

>(p.s., may I suggest - seriously - that if the doctors and wanna-be-doctors on
>the net who refuse to have an open mind on alternative treatments and
>theories, such as the "yeast theory", should create your own moderated group.

Why? Is there some reason why you feel that it shouldn't be pointed out
in SCI.med that there is no convincing empirical evidence to support the
existence of systemic yeast syndrome?

--
David Rind
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